

Project Design Phase
Proposed Solution

Date	18th June 2026
Student	Tirumala Srindhi
Project Name	ShopEZ – E-Commerce Application
Maximum Marks	2 Marks

Proposed Solution Template:

Client (Frontend)

- **Tech Stack:** HTML, CSS, JavaScript, React.js
- **Key Components:**
 - User Authentication (Login/Register)
 - Product Catalog (with “Shop Now” button)
 - Cart Management
 - Order Details Page
 - Secure Checkout Flow
 - Profile Section (orders, saved details)
 - Admin Dashboard (product/category management)

Server (Backend)

- **Tech Stack:** Node.js, Express.js
- **API Endpoints:**
 - Api/users → Register/Login/Profile
 - Api/products → Product listing, search, categories
 - Api/cart → Add/remove items, view cart
 - Api/orders → Place order, view order history
 - Api/admin → Admin authentication, dashboard

Database

- **MongoDB Collections:**
 - Users → User details, credentials
 - Products → Product catalog with categories
 - Cart → Items linked by user ID
 - Orders → Order history, payment, shipping details
 - Admin → Banner images, categories

S.No.	Parameter	Description
1.	Problem Statement (Problem to be solved)	Lack of a seamless, secure, and user-friendly e-commerce platform for both buyers and sellers.
2.	Idea / Solution description	Shop-EZ provides a one-stop online shopping solution with intuitive navigation, secure checkout, and seller analytics.
3.	Novelty / Uniqueness	Combines effortless shopping with seller dashboards and real-time analytics.
4.	Social Impact / Customer Satisfaction	Enhances customer trust with secure transactions and boosts seller growth with insights.
5.	Business Model (Revenue Model)	Commission-based revenue on sales + premium seller subscriptions.
6.	Scalability of the Solution	Easily scalable with microservices and cloud deployment; supports large product catalogs and multiple sellers.